

S-parameters for 2-port networks

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What is a network?

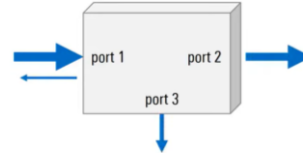
- In the RF world, a network is a device with one or more ports
- Each port can pass, absorb, and/or reflect RF energy
- Examples
 - 1 port: Antenna, Load (or dummy load)
 - 2 port: Filter, Amplifier, Attenuator transmission line
 - 3 port: Directional Coupler, Mixer

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Analyzing networks

- Networks can be analyzed by:
 1. Injecting RF energy into a given port
 2. Measuring the level of RF appearing
 - At that port (reflected)
 - At other ports



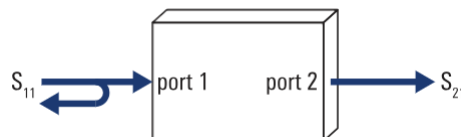
- Usually only one signal is injected into one port at one time (and match the other ports)
- Usually measured over a range of frequencies to characterize frequency specific behaviors
- Networks are typically analyzed using an instrument called a network analyzer

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What are S-parameters?

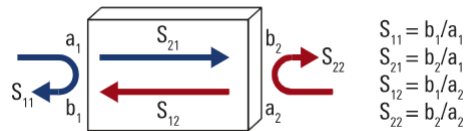
- S-parameters are the most common way to represent these measurements
- S= Scattering
 - Since we are measuring how much energy is scattered back from and out of the various ports
- Names using the letter 'S' and pair of subscripts S_{xy}
 - x: port where the energy emerges (output port)
 - y: port where the energy enters (input port)



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S-parameters for 2-port network



- **S_{11} is the input reflection coefficient.** It measures the portion of the input signal at port 1 that is reflected back to port 1. This parameter indicates how much of the input signal is reflected due to an impedance mismatch.
- **S_{21} is the forward transmission coefficient.** It measures the portion of the input signal at port 1 that is transmitted to port 2. This parameter indicates the efficiency of signal transmission from the input to the output.
- **S_{12} is the reverse transmission coefficient.** It measures the portion of the input signal at port 2 that is transmitted to port 1. This parameter indicates how well port 1 is isolated from signals entering port 2.
- **S_{22} is the output reflection coefficient.** It measures the portion of the input signal at port 2 that is reflected back to port 2. This parameter indicates how much of the output signal is reflected due to impedance mismatch at the output.

When Port 2 is matched

When Port 1 is matched

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S-parameters

For an N -port network, the S-parameters will be represented by an $N \times N$ matrix.

For 1-port:

$$S = [S_{11}] \quad \text{which is equal to } \Gamma$$

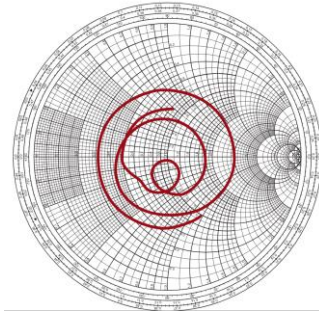
For a 2-port network:

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

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More about S-parameters

- S-parameters are complex values with:
 - Magnitude
 - Phase
- Reflection coefficients (S_{xx}) can be plotted on a Smith chart
- Common names:
 - Reflection Coefficients
 - S_{11} : input match, return loss, VSWR
 - S_{22} : output match, return loss, VSWR
 - Transmission Coefficients
 - S_{21} : gain or loss or insertion loss
 - S_{12} : reverse isolation “how much power is flowing the wrong way”

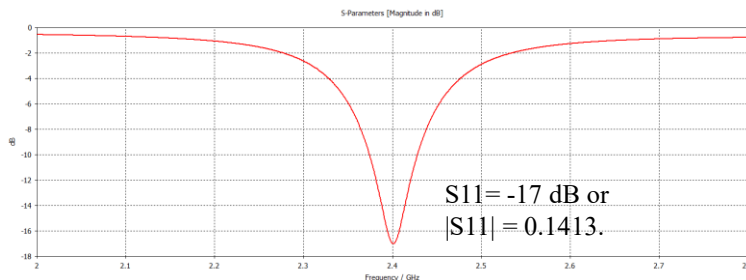


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Example of [S]

- 1-port
 - Ideal termination (matched): $S_{11} = 0$, or ideal short: $S_{11} = -1$
 - Antenna
 - $RL = -20 \log_{10} |S_{11}|$



If $S_{11} = 0 \text{ dB}$, then all the power is reflected from the antenna, and nothing is radiated.

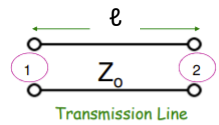
If $S_{11} < -10 \text{ dB}$, this implies most of the power is “accepted by” or delivered to the antenna. This accepted power is either radiated or absorbed as losses within the antenna.

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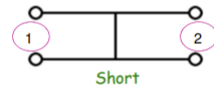
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Example of [S]

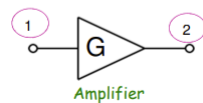
- 2-port



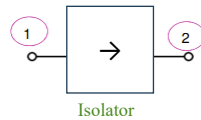
$$[S] = \begin{bmatrix} 0 & e^{-j\beta\ell} \\ e^{-j\beta\ell} & 0 \end{bmatrix}$$



$$[S] = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



$$[S] = \begin{bmatrix} 0 & 0 \\ G & 0 \end{bmatrix}$$



$$[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

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2-Port Network properties

- If network is lossless (it satisfies unitary condition):

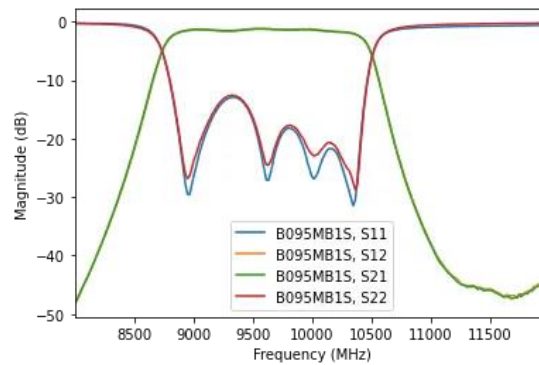
$$[S]_t [S]^* = [U]$$

- If network is reciprocal:
 - $S_{12} = S_{21}$
 - Example of non-reciprocal are direction dependent devices
Amplifiers, isolators (use anisotropic materials like ferrites)
- If network is symmetrical:
 - $[S] = [S]_t$ or $S_{12} = S_{21}$ & $S_{11} = S_{22}$
 - Passive and isotropic (e.g. cables and attenuators)

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Bandpass filter S-parameters



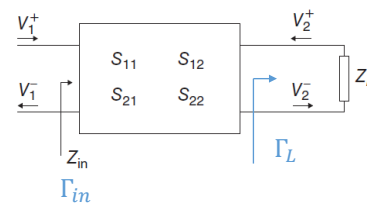
- $RL = -20 \log_{10}|S_{11}|$
- $IL = -20 \log_{10}|S_{21}|$

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Analysis of 2-port network

- 2-port terminated with a load



$$\Gamma_{in} = S_{11} + \frac{S_{12}\Gamma S_{21}}{1 - \Gamma S_{22}}$$

$$Z_{in} = \frac{1 + \Gamma_{in}}{1 - \Gamma_{in}}$$

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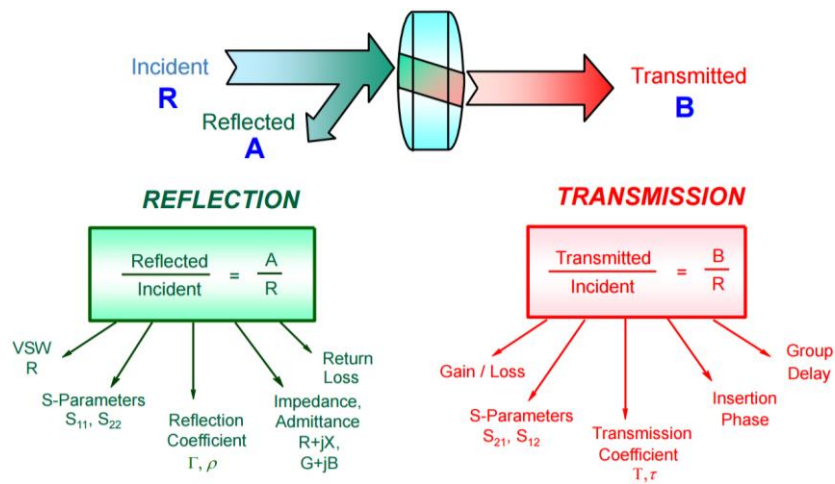
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Idea behind Vector Network Analyzer (VNA)

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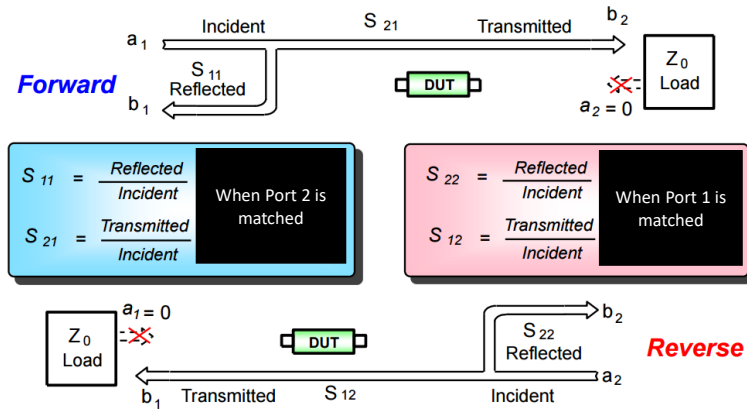
High Frequency Device Characterization



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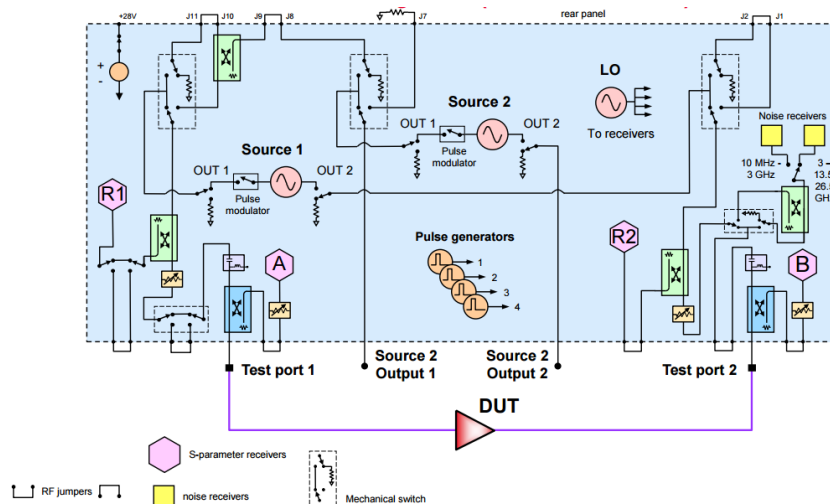
S-Parameters



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VNA Block Diagram



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Different VNA's



Benchtop VNA



Handheld VNA



"USB" VNA



Nano VNA

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VNA Front Panel



VNA: N9918A

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Why do we need Calibration?



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Introduction – Why Calibration?



Systematic errors

- Due to **imperfections** in the analyzer and test setup
- Assumed to be **time invariant** (predictable)
- Generally, are largest sources of error



Random errors

- **Vary** with time in random fashion (unpredictable)
- Main contributors: instrument **noise**, switch and connector **repeatability**



Drift errors

- Due to system performance changing **after** a calibration has been done
- Primarily caused by **temperature variation**



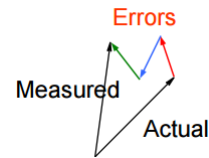
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Introduction – How to Calibrate?

Vector-error correction

- Is a process for characterizing systematic error terms
- Measures **known electrical standards**
- Removes effects of error terms from subsequent measurements



Electrical Standards (Cal kits)

- Can be **mechanical** or **electronic**
- Are often an **Short, Open, Load, and Thru**, but can be arbitrary impedances as well



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1 - Attach the cables



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2- Set your frequency range



300 kHz to 3 GHz

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3- Choose Calibration Kit Model

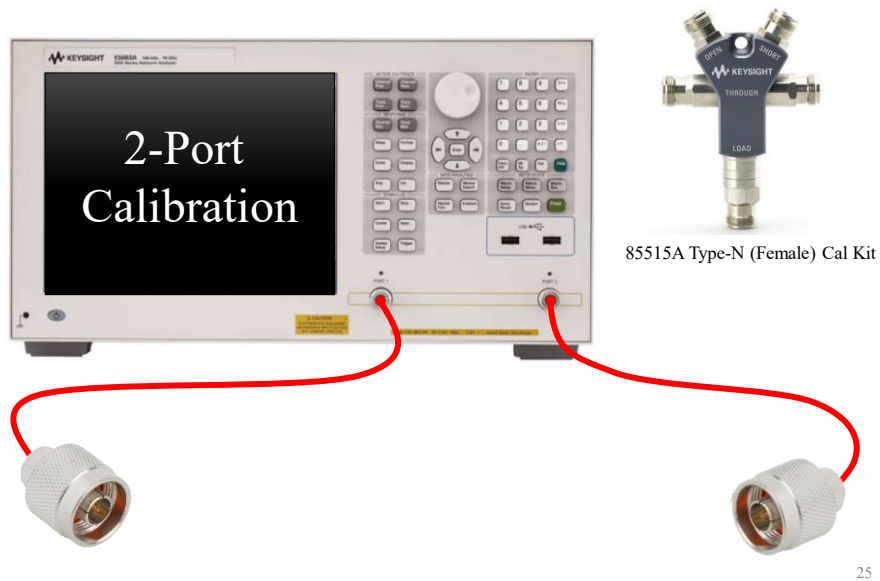


85515A Type-N (Female) Cal Kit

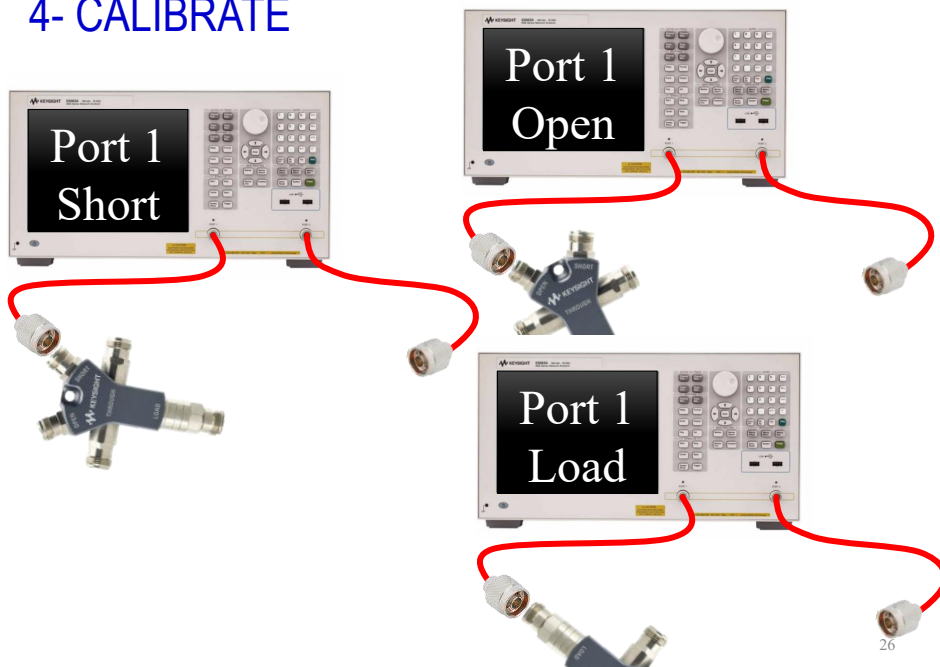
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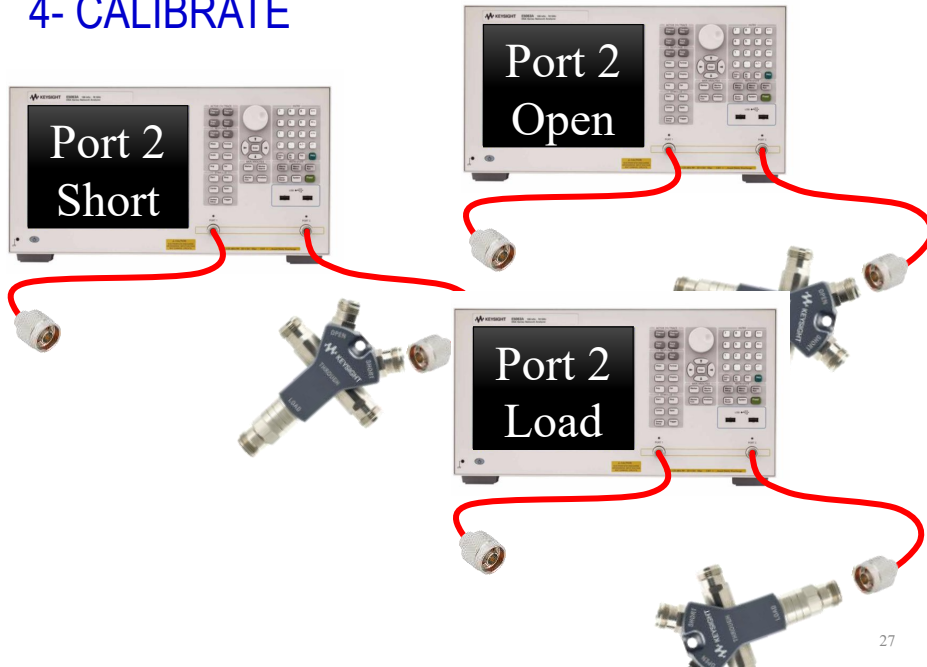
4- Choose 1-port Cal OR 2-port Cal



4- CALIBRATE



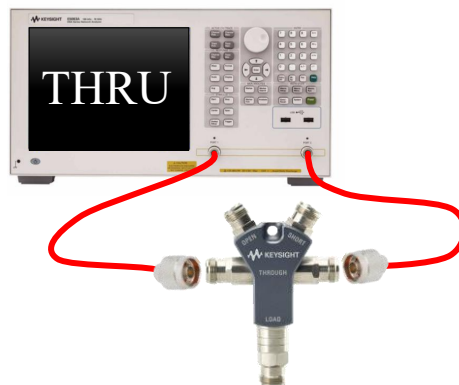
4- CALIBRATE



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4- CALIBRATE

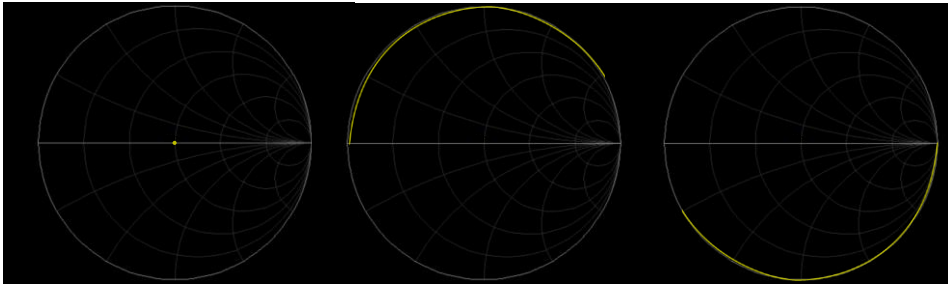


Finally press Done
A "beeping" sound will indicate calibration was done successfully

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5- Test the calibration



P	Load	SC	OC
1	✓	✓	✓
2	✓	✓	✓



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Calibration Demo

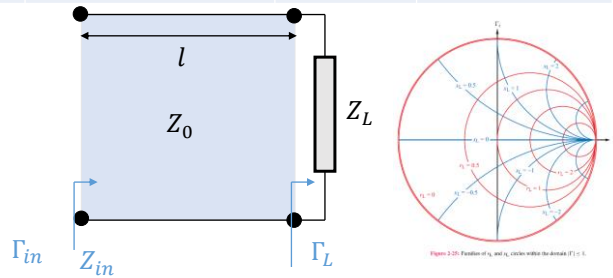
- Watch VNA Calibration video [playlist](#)

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NEXT LECTURE: Smith Charts!

	Z_L	Z_{in}	Γ	VSWR
Load Z_L	Z_L	$= Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$	$= \frac{Z_L - Z_0}{Z_L + Z_0}$	$= \frac{1 + \Gamma }{1 - \Gamma }$
Matched (if $Z_0 = 50\Omega$)	50Ω	$= Z_0$	0	1
Open Circuit (OC)	∞	$= -jZ_0 \cot(\beta l)$	+1	∞
Short Circuit (SC)	0Ω	$= jZ_0 \tan(\beta l)$	-1	∞



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